

# ManoVarta Milestone 2: Baseline Report for Multilingual Conversational Mental Health Screening

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## Abstract

This report presents the **Milestone 2 baseline** for ManoVarta, a multilingual conversational screening assistant for mental health. The system converts open-ended English, Hindi, and Hinglish dialogue into item-level PHQ-9/GAD-7 evidence and 0–3 severity scores, with a separate safety label (none/review/urgent). We evaluate the milestone model trail used in this phase: heuristic seed runtime, direct Aya/Kimi/Qwen prompting, supplementary local open-model probes (Qwen2.5-1.5B and Llama 3.1 8B on a small  $n=3$  slice), structured evidence-first Aya extraction, DAIC-auxiliary continuation, and an exploratory hybrid safety pilot. The key baseline finding is that conversational fluency does not imply screening completeness: heuristic/direct baselines are sparse or invalid, while evidence-first extraction substantially increases coverage (0.913), and DAIC auxiliary continuation further clarifies the safety-coverage trade-off.

## 1 Problem Statement

Clinical screening tools such as PHQ-9 and GAD-7 are reliable and interpretable, but rigid administration often feels burdensome in real dialogue (Kroenke et al., 2001; Spitzer et al., 2006). Our goal is to build a screening partner that keeps conversation natural while preserving structured outputs needed for downstream triage and review (Abd-Alrazaq et al., 2020; Dosovitsky et al., 2021).

For Milestone 2, the baseline objective is to show a working direction:

- multilingual open-ended chat intake (English/Hindi/Hinglish),
- item-level PHQ-9/GAD-7 extraction and scoring,

| Dataset                  | Train | Dev | Test |
|--------------------------|-------|-----|------|
| Extractor core           | 84    | 48  | 48   |
| Extractor DAIC-augmented | 191   | –   | –    |
| Follow-up planner        | 120   | 96  | 144  |
| Safety classifier        | 126   | 80  | 84   |

Table 1: Milestone-2 data slices used for baseline experiments.

- confidence/coverage-driven follow-up behavior,
- explicit safety trigger pathway.

## 2 Data

### 2.1 Data Used in This Milestone

The baseline pipeline uses transcript-centered multilingual training/evaluation data under data/processed/. The extractor held-out split is language-balanced. We also keep a DAIC-augmented extractor training slice as an auxiliary English-oriented supervision source, while treating it cautiously for conversational generalization (Burdisso et al., 2024). Extractor language distribution is balanced in held-out data: Dev/Test each contain 16 English, 16 Hindi, and 16 Hinglish examples. Multilingual handling and code-mixed robustness decisions follow prior Indic NLP and Hinglish modeling evidence (Kakwani et al., 2020; Nayak and Joshi, 2022).

**Data-status scope.** In this milestone report, labels are treated as engineering-time supervision for baseline comparison and model shaping. We do not present this milestone as psychiatrist-certified final clinical labeling.

**Scope boundary against full report.** The reference project PDF includes later deployment/runtime stages. Those are intentionally excluded here. This document is restricted to

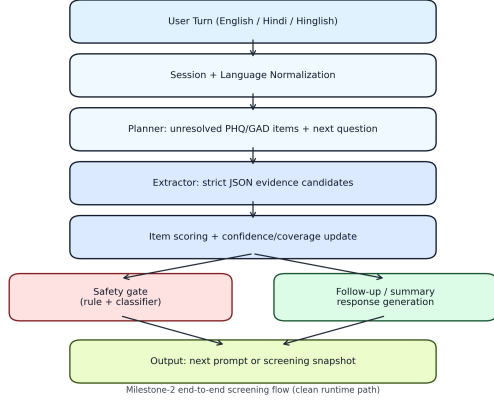


Figure 1: End-to-end milestone baseline pipeline from user query to structured output.

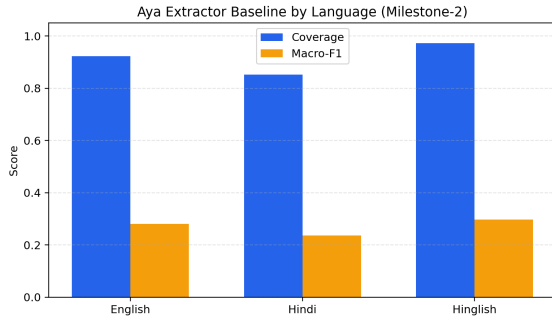


Figure 2: Language-wise Aya extractor baseline performance in Milestone 2.

Milestone-2 baseline systems, probes, and their measured trade-offs.

## 2.2 Sample Transcript Excerpts

## 3 Baseline Architecture

The baseline runtime flow is shown in Figure 1: ingest user turn, update dialogue state, run planner, run extractor, score PHQ/GAD items, run safety trigger path, and return follow-up or screening snapshot.

**Confidence proxy (milestone definition).** We use coverage completeness as an operational confidence proxy:

$$\text{coverage completeness} = \frac{\text{covered evaluable items}}{\text{total evaluable items}}.$$

Uncovered items are treated as unresolved and prioritized in subsequent follow-up turns.

### 3.1 Model and System Configurations

The milestone model sweep included Aya-family prompting (Cohere For AI, 2024) and Qwen-

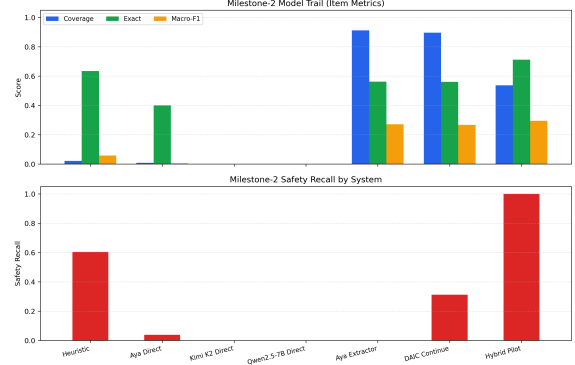


Figure 3: Milestone-2 baseline metrics across full held-out systems and supplementary experiments.

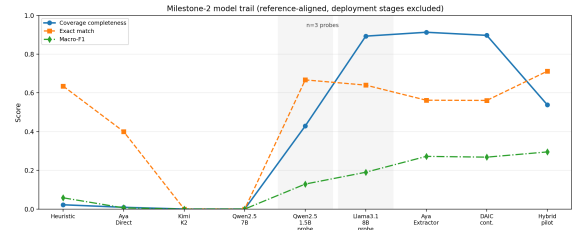


Figure 4: Model trail for Milestone 2 only. Grey bands indicate supplementary  $n=3$  probes.

family prompting (Qwen Team, 2026). We include local Qwen/Llama probe rows for completeness because they influenced model-selection discussion, even though both are small-slice checks and not full held-out multilingual benchmarks. Mistral NeMo and Gemma appeared in planning notes, but we do not report fabricated numbers because no archived reproducible milestone result files were found (Mistral AI, 2026; Google DeepMind, 2026).

## 4 Experimental Setup

Evaluation metrics. We report:

- **Coverage completeness:** how much of the required item space is touched,
- **Exact match** and **macro-F1:** item-level scoring quality,
- **MAE:** severity error magnitude,
- **Safety precision/recall:** escalation quality.

Safety is separately tracked because conversational screening can appear coherent while missing risk cues (Zirikly et al., 2019; Pichowicz et al., 2025).

| Language | Excerpt                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| English  | <b>assistant:</b> Over the last couple of weeks, what has felt the heaviest lately?<br><b>user:</b> Mostly I feel drained all day. My sleep schedule is messed up and I keep worrying that I am falling behind.<br><b>assistant:</b> When you say falling behind, is it more focus, mood, or worry?<br><b>user:</b> Focus for sure. I stare at the screen and nothing sticks, and the worry does not really stop. |
| Hindi    | <b>assistant:</b> Pichhle do hafton mein sabse zyada kis baat ne pareshan kiya?<br><b>user:</b> Dimag hamesha chalta rehta hai. Raat ko bhi soch band nahi hoti.<br><b>assistant:</b> Kya is wajah se neend ya aaraam par bhi asar pada hai?<br><b>user:</b> Haan, neend toot toot ke aati hai aur subah se hi ghabrahat rehti hai.                                                                               |

Table 2: Representative multilingual excerpts from the milestone extractor dataset.

| System                        | Core model/runtime                                   | Output form                      | Setup notes                                                                |
|-------------------------------|------------------------------------------------------|----------------------------------|----------------------------------------------------------------------------|
| Heuristic seed baseline       | Rules + lexicons + question-naire map                | Item snapshot + safety label     | No generative model in scoring loop                                        |
| Direct LLM baseline           | CohereLabs/aya-expanse-32b                           | Direct item JSON from raw prompt | Single-pass direct scoring                                                 |
| Direct LLM baseline           | Kimi K2 endpoint (prompt-only)                       | Direct item JSON from raw prompt | Produced no valid evaluable outputs in milestone test pass                 |
| Direct LLM baseline           | Qwen2.5-7B endpoint (prompt-only)                    | Direct item JSON from raw prompt | Produced no valid evaluable outputs in milestone test pass                 |
| Supplementary probe           | Local Qwen2.5-1.5B checkpoint                        | Direct item JSON from raw prompt | Small audit slice ( $n=3$ ); one Hindi parse failure                       |
| Supplementary probe           | Local Llama 3.1 8B quantized probe                   | Direct item JSON from raw prompt | Retrospective small audit slice ( $n=3$ ), safety over-escalation observed |
| Structured extractor baseline | CohereLabs/aya-expanse-32b                           | Strict JSON: items, safety_level | Evidence-first extraction with schema constraints                          |
| DAIC auxiliary continuation   | Aya continuation with DAIC-WOZ auxiliary supervision | Item JSON + safety               | Coverage remained high, but safety recall stayed limited                   |
| Exploratory hybrid pilot      | Local IndicBERT safety + rules                       | none/review/urgent               | Safety fusion pilot on runtime path                                        |

Table 3: System variants tracked in Milestone 2, including supplementary probe checkpoints.

## 5 Results

**Evaluation criteria coverage (explicit).** To match the project rubric dimensions, we explicitly include: **Disclosure efficiency** = 2.292 average turns and 2.0 median turns to stable item score (30 conversations, 96 item traces); **Latency** = 937.69 ms cold start, 29.55 ms warm average, 32.73 ms warm median, and 32.89 ms warm p95 over 6 samples.

**Language-wise extractor baseline.** The evidence-first Aya extractor baseline is multilingual but uneven: English (coverage 0.922, exact 0.593, macro-F1 0.281), Hindi (0.852, 0.486, 0.236), and Hinglish (0.973, 0.606, 0.297).

**Safety checkpoint pilot.** Standalone local safety checkpoint (84 examples): 0.6548 accuracy, 0.4733 macro-F1; gold label counts = 32 none / 40 review / 12 urgent, predicted counts = 59 none / 25 review / 0 urgent.

## 6 Failure Analysis from Long-Conversation Stress Tests

During manual long-turn testing in English, Hindi, and Hinglish, we repeatedly observed:

- **Repetitive closure loops:** near-identical summary/closure responses over many turns,
- **Planner over-structure:** responses felt checklist-like rather than natural,
- **Follow-up drift:** after apparent closure, prompts reverted to generic follow-ups.

## 7 Limitations and Milestone Scope

This is a **Milestone-2 baseline report**, not a final system claim. Key limitations:

- structured extraction improves coverage but baseline safety is weak without hybrid controls,

- multilingual calibration remains uneven (Hindi weakest among baseline language slices),
- long-conversation naturalness degrades due to closure-loop behavior,
- probe rows and DAIC continuation are separated as diagnostics for transparency, not primary held-out endpoints.

## 8 Conclusion

Milestone 2 establishes the right direction: evidence-first extraction is necessary for multilingual conversational screening, and safety/discourse robustness require dedicated controls beyond direct prompting.

**Artifact traceability.** All key numbers are drawn from: [reports/acl\\_paper/metrics\\_snapshot.json](#), [reports/aya\\_colab\\_eval\\_a100\\_20260328.json](#), [reports/colab\\_daic\\_continue\\_vm\\_20260413/extractor\\_eval\\_full\\_rerun/summary.json](#), and [reports/colab\\_daic\\_continue\\_vm\\_20260413/eval\\_full\\_localbase\\_20260413-042649.log](#).

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| Main held-out systems (Milestone 2) | Coverage | Exact | Macro-F1 | MAE   | Safety P/R    |
|-------------------------------------|----------|-------|----------|-------|---------------|
| Heuristic seed runtime              | 0.022    | 0.635 | 0.058    | 0.492 | 1.000 / 0.604 |
| Direct Aya prompting                | 0.009    | 0.400 | 0.006    | 0.840 | 0.667 / 0.038 |
| Direct Kimi K2 prompting            | 0.000    | –     | 0.000    | –     | 0.000 / 0.000 |
| Direct Qwen2.5-7B prompting         | 0.000    | –     | 0.000    | –     | 0.000 / 0.000 |
| Structured Aya extractor baseline   | 0.913    | 0.562 | 0.272    | 0.443 | 0.000 / 0.000 |
| Exploratory hybrid safety pilot     | 0.538    | 0.712 | 0.295    | 0.288 | 1.000 / 1.000 |

Table 4: Primary held-out Milestone-2 results. “–” means no valid evaluable outputs were produced for that metric in the archived run.

| Diagnostics / supporting runs                                                                         | Evidence type                                                       |
|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Local Qwen2.5-1.5B probe ( $n=3$ ): cov 0.429, exact 0.667, F1 0.129                                  | Small-slice local probe; not a full held-out benchmark              |
| Retrospective Llama 3.1 8B probe ( $n=3$ ): cov 0.893, exact 0.640, F1 0.190                          | Retrospective small-slice probe; not used as main baseline          |
| DAIC auxiliary continuation (v2run3): cov 0.897, exact 0.561, F1 0.268, MAE 0.458, safety 1.000/0.312 | Archived continuation logs/summary; auxiliary continuation evidence |

Table 5: Separated diagnostics to improve reproducibility clarity.